



江苏省苏州中学国际部
Suzhou High School – International Division

SZZX-ID Bridge Program Physics

Unit 3 Lesson 3: circular motion – applications

Class Outline

Title of Unit	Projectile and circular motions
Title of Lesson	circular motion - applications
Starter Activity	Find circular motions around us
Direct Instructions	Teaching Strategies: Demonstration, Lecture, Collaboration, Questioning, Inquiry
	Lecture: Examples of circular motions. Application of centrifugal effect. Collaboration and Inquiry: explain why we need to drive carefully in rainy days, on both straight road and when turning
	Differentiation: Content, Instructional Strategies, the Classroom, Products, Teacher
	Allow discussion in Chinese in Group discussion. Encourage using English and physical terminologies to explain
	International Mindedness:
	The physics behind road safety
Lesson Objectives	By the end of the lesson, students will be able to: - Understand the examples and applications of circular motion in real life - Know how to apply the formulas of circular motion in real examples
ATLs	Identify unit ATLs and how they are being addressed in the lesson
	Communication Social
IB Learner Profile	Link the lesson to the characteristics of the IB Learner attribute
	thinker
Teaching Activities	Group activity: explain why we need to drive carefully in rainy days, on both straight road and when turning Video: velocity of circular motion, acceleration of circular motion
Plenary Session	Checking for understanding / Homework / Q & A / Reflection etc.
	Homework instruction – group work

Vocabulary



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- Banked 倾斜的
- Bird's eye view 俯视图
- Slide 滑
- Tire 轮胎
- Fictitious force 虚拟力
- Artificial 人工的/假的

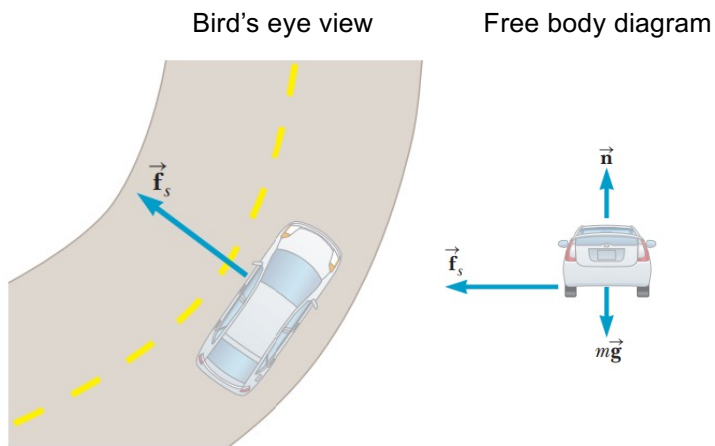
Making a Turn



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A car travels at a constant speed of 13.4 m/s on a circular turn of radius 50.0 m, as shown in the bird's-eye view in the figure.

What minimum coefficient of static friction, μ_s , between the tires and roadway will allow the car to make the circular turn without sliding?



- The centripetal force is provided by the force of static friction, which is directed radially toward the center of the circular path
- Calculating the force of friction requires normal force. In this case from Newton's second Law, $N = mg$

From the expression of centripetal force: $f_{s,max} = m \frac{v^2}{r}$

For max. static friction: $f_{s,max} = \mu_s N$, where $N = mg$

$$\mu_s mg = m \frac{v^2}{r} \quad \Rightarrow \quad \mu_s = \frac{v^2}{rg} = \frac{13.4^2}{50 \times 9.81} = 0.366$$

- What will happen if speed exceeds 13.4m/s in the case above?
- If μ_s further decrease, would the safe speed limit increase, decrease or remain the same?

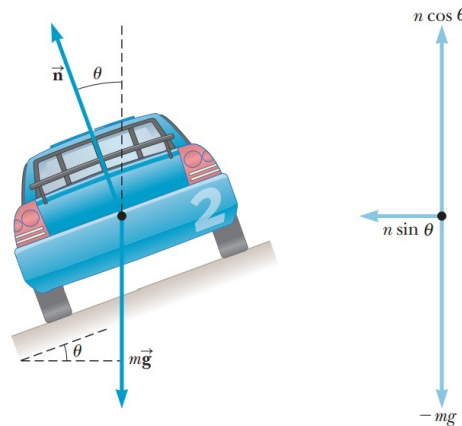
Daytona 500 banked curve



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The Daytona International Speedway in Daytona Beach, Florida, is famous for its races, especially the Daytona 500, held every February. Its track features a 31.0° banked curve, with a maximum radius of 316 m. If a car passes the curve too slowly, it tends to slip down the incline of the turn, whereas if it's going too fast, it may begin to slide up the incline.

- (a) Find the necessary centripetal acceleration on this banked curve so the car won't slip down or slide up the incline. (Neglect friction for now.)
(b) Calculate the speed of the race car.



a). Vertically: $N \cos \theta = mg \Rightarrow N = \frac{mg}{\cos \theta}$

For the circular motion: $N \sin \theta = ma$ Where a is the centripetal acceleration

$$a = \frac{N \sin \theta}{m} = g \frac{\sin \theta}{\cos \theta} = g \tan \theta = 9.81 \times \tan 31^\circ = 5.89 \text{ m/s}^2$$

b). $a = \frac{v^2}{r} \Rightarrow v = \sqrt{ar} = \sqrt{5.89 \times 316} = 43.1 \text{ m/s}$

What will happen when velocity is higher than this?
Lower than this?

The Mysterious Centrifugal Force



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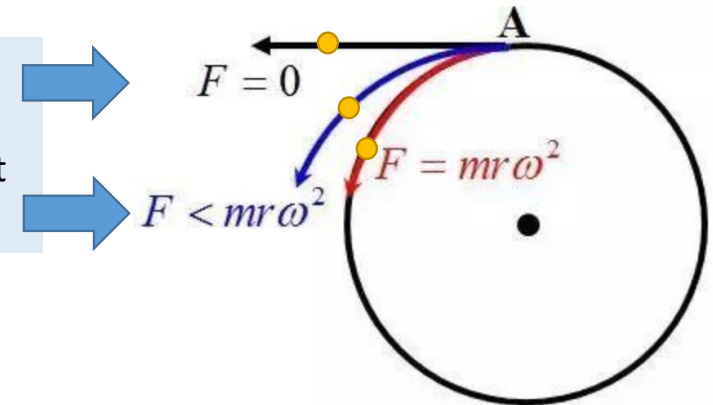
For an object on circular motion, at point A,

1. if the centripetal force continues, it will continue to move in a circular motion (red track)
2. If the centripetal force disappears, the object will move with the same velocity, along the tangent at A (black track)
3. What if the force is between the case of 1 and 2? The motion also follow a path between red track and black track.

Centrifugal Force

A so-called centrifugal force is most often just the absence of an adequate centripetal force. It is a fictitious force.

It may look like someone is applying a force on the yellow ball and pulling it out from the circular motion.



The Mysterious Centrifugal Force



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The “centrifugal force” or centrifugal effect is widely used in life



Centrifugal separation



Spin cycle of a washing machine



Artificial gravity?

Group work – let's explain the physics!



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Explain with physics principles why we need to drive more slowly and more carefully on rainy days

- on a straight road. or
- when making a turn.

◆ Communication: Organize and depict information logically

- Try this sequence: Law – equation – analysis - conclusion
- Which law/equations of physics can I apply? Why?
- Which day-to-day language term I should translate to physics languages in order to apply the formulas?
- Which physics languages should I translate to day-to-day language in order to give my final explanation?

◆ Social: Working collaboratively to achieve a common goal

Explain to your teammates and hear their comments.

- Submission: a set of handwritten/drawn explanations. Use diagrams and formulas to explain whenever possible